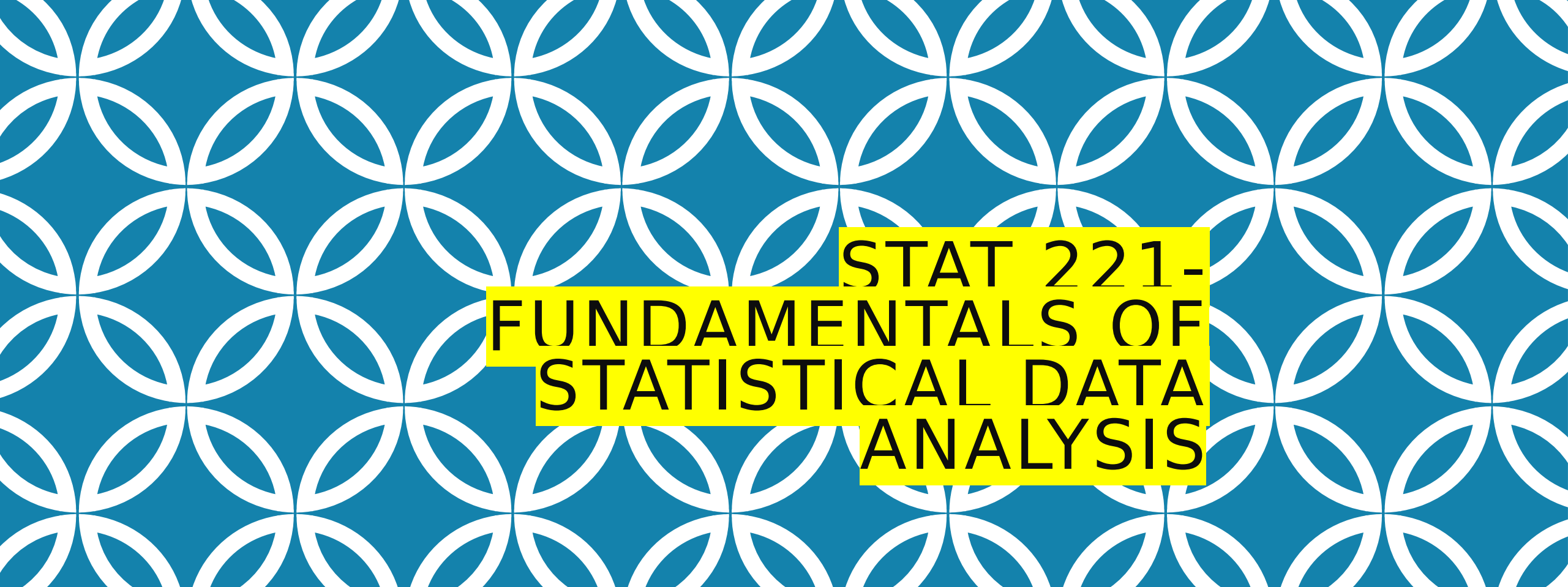
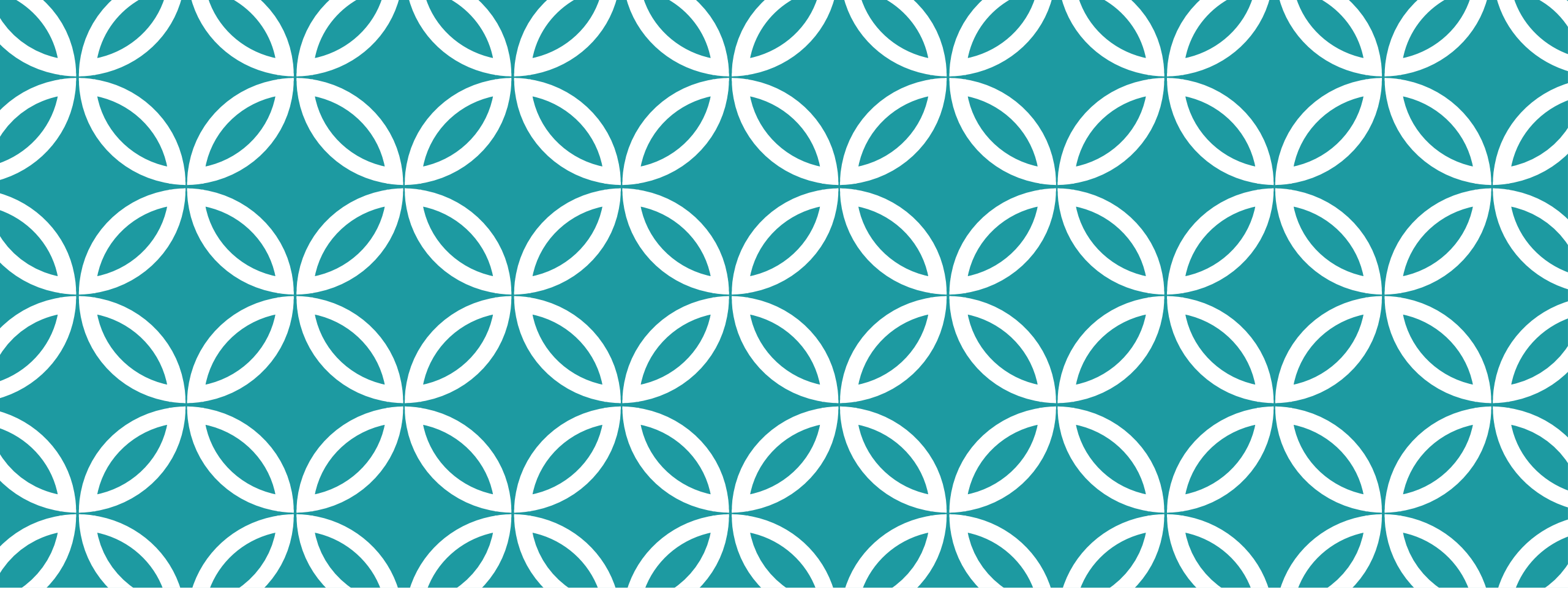




STAT 221- FUNDAMENTALS OF STATISTICAL DATA ANALYSIS

Week 10- Probability for social science -1

Shivakumar Jolad



PROBABILITY FOR SOCIAL SCIENCE |

SECTIONS

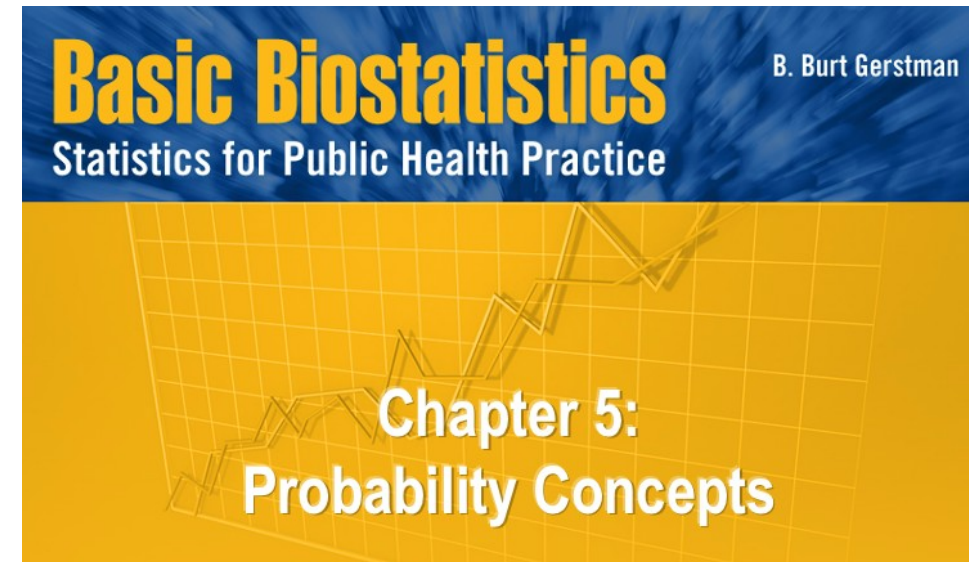
What is Probability?

Types of Random Variables

Discrete Random Variables

Continuous Random Variables

More Rules and Properties of Probability



DEFINITIONS

Random variable $\equiv$ a numerical quantity that takes on different values depending on chance

Population $\equiv$ the set of all possible values for a random variable

Event $\equiv$ an outcome or set of outcomes

Probability $\equiv$ the relative frequency of an event in the *population* ... alternatively... **the proportion of times an event is *expected* to occur in the long run**

EXAMPLE

In a given year: 42,636 traffic fatalities (events) in a population of $N = 293,655,000$

Random sample population

Probability of event
= relative freq in pop
= $42,636 / 293,655,000$
= .0001452
 ≈ 1 in 6887



BASIC PROBABILITY



<https://www.youtube.com/watch?v=iLaSpfgVmCA>

QUANTIFYING UNCERTAINTY

Probability is
used to
quantify levels
of belief:

Probability	Expression
0.00	Never
0.05	Seldom
0.20	Infrequent
0.50	As often as not
0.80	Very frequent
0.95	Highly likely
1.00	Always

RANDOM VARIABLES

Random variable $\equiv$ a numerical quantity that takes on different values depending on chance

Two types of random variables

- ▮ **Discrete random variables** (countable set of possible outcomes)

- ▮ **Continuous random variable** (unbroken chain of possible outcomes)

EXAMPLE: DISCRETE RANDOM VARIABLE

Treat 4 patients with a drug that is 75% effective

Let $X \equiv$ the [variable] number of patients that respond to treatment

X is a discrete random variable can be either 0, 1, 2, 3, or 4 (a countable set of possible

Social science example:

In a population – 75% literacy ;

If you randomly pick a sample of 4 people

X = no of people who are literate

$X = \{0, 1, 2, 3, 4\}$

Chance that a random person literate? $3/4$



EXAMPLE: DISCRETE RANDOM VARIABLE

Discrete random variables are understood in terms of their **probability mass function (*pmf*)**

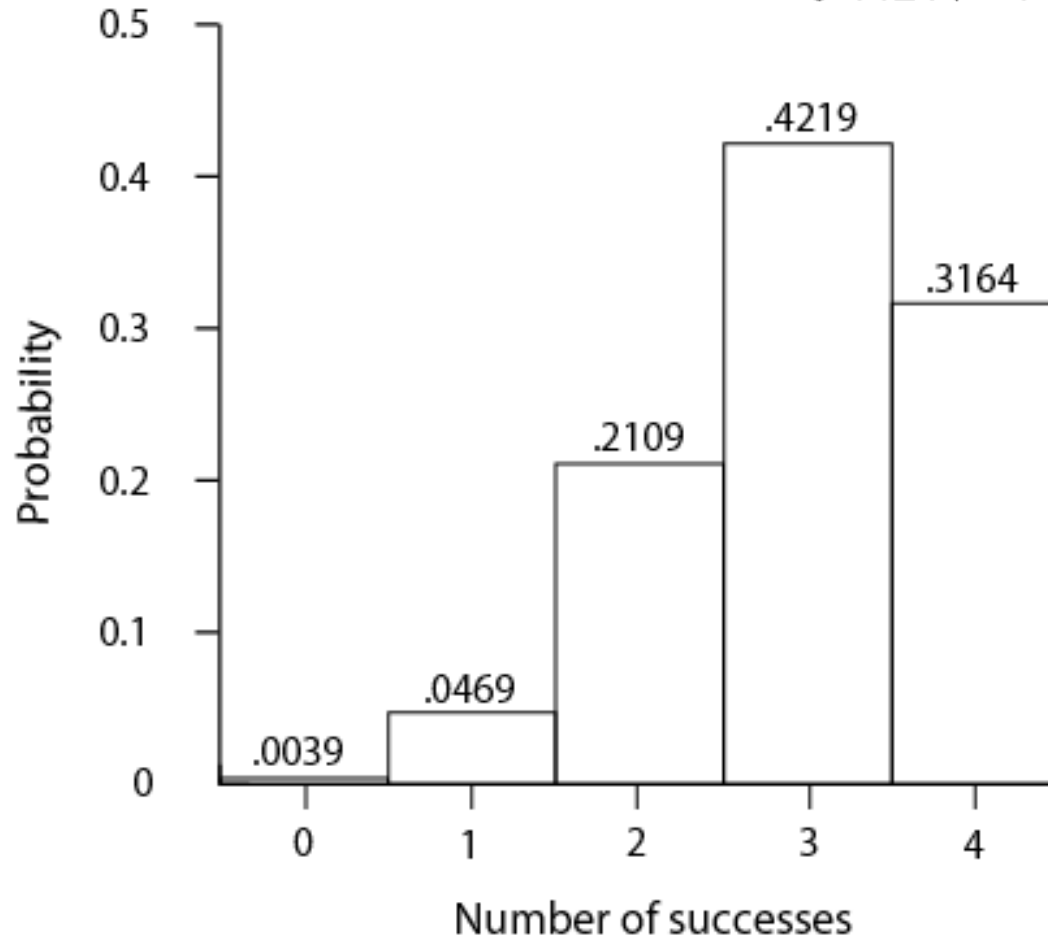
pmf $\equiv$ a mathematical function that assigns probabilities to all possible outcomes for a discrete random variable.

This table shows the *pmf* for our “four people” example:

x	0 (none literate)	1 (one literate)	2 (two are literate)	3 (three are literate)	4 (All four are literate)
$\Pr(X=x)$	0.0039	0.0469	0.2109	0.4219	0.3164

PMF CAN ALSO BE SHOWN GRAPHICALLY

$$\mu = \text{Mean} = 0 * 0.0039 + 1 * 0.0469 + 2 * 0.2109 + 3 * .4219 + 4 * .3164 = 3$$



$$\sum_x p(x) * (x - \mu)^2$$

AREA ON *PMF* = PROBABILITY

Areas under *pmf* graphs correspond to probability

For example:

$\Pr(X = 2)$

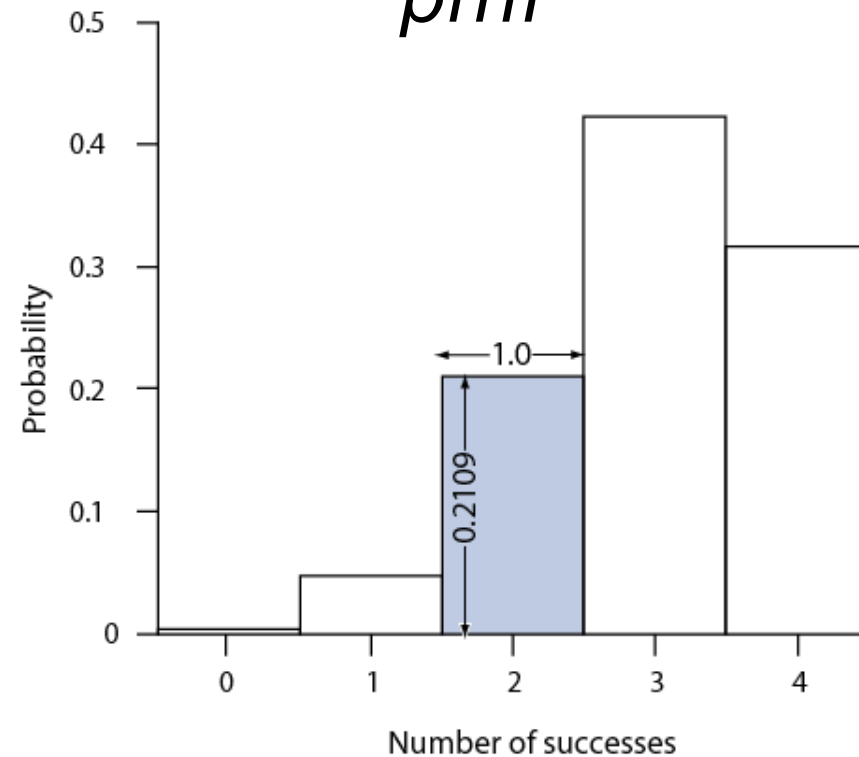
= shaded rectangle

= height $\times$ base

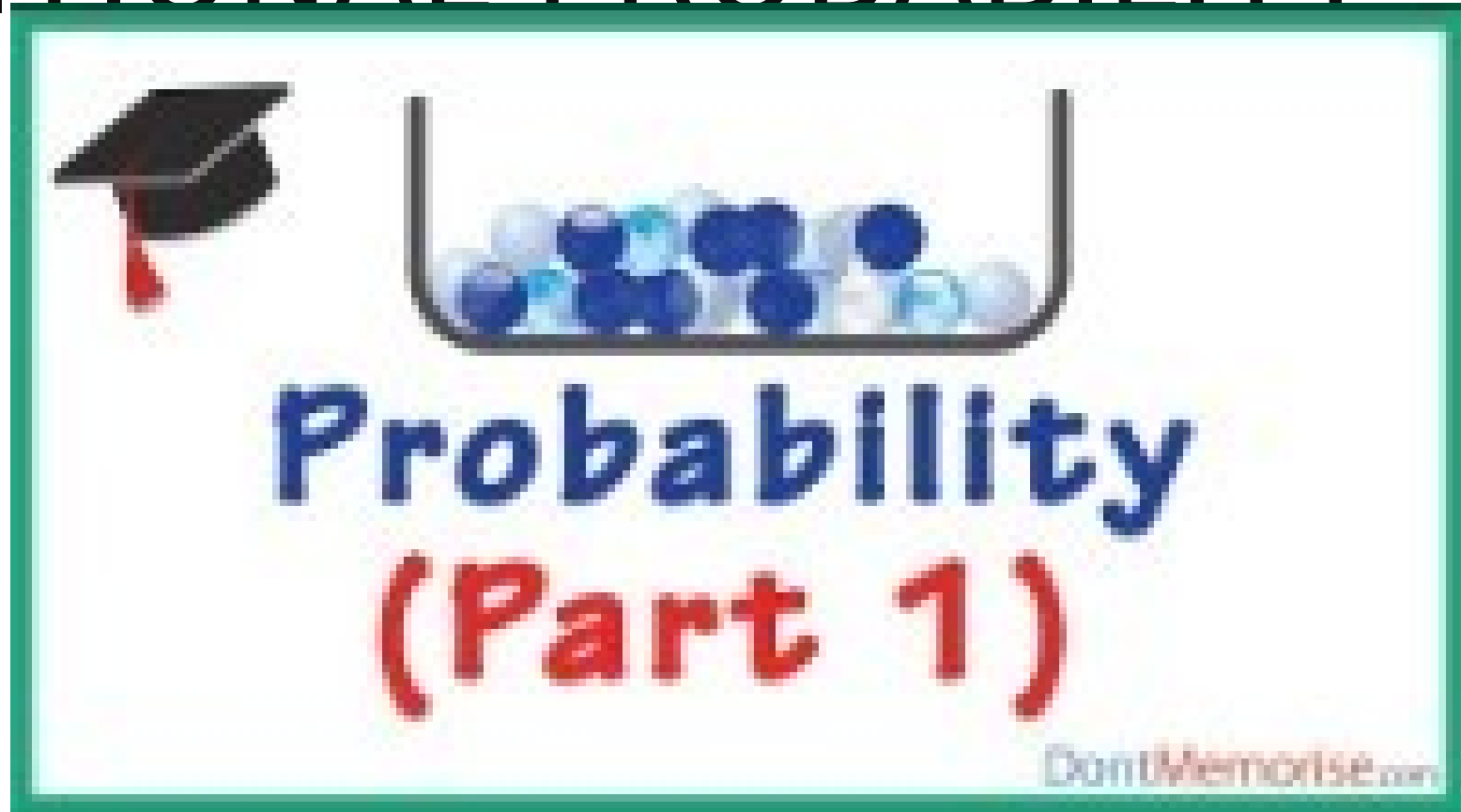
= $.2109 \times 1.0$

= $.2109$

“Four patients”
pmf



PROBABILITY- MULTIPLICATION RULE + CONDITIONAL PROBABILITY



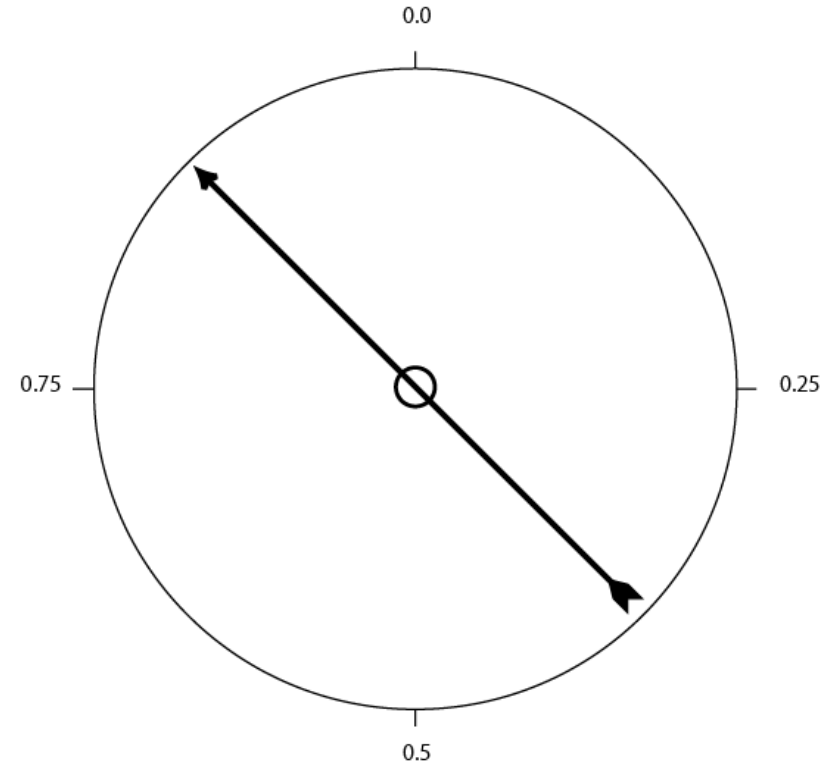
<https://www.youtube.com/watch?v=8zMuKn2qORA>

EXAMPLE: CONTINUOUS RANDOM VARIABLE

Continuous random variables have an infinite set of possible outcomes

Example: generate random numbers with this spinner $\Rightarrow$

Outcomes form a *continuum* between 0 and 1

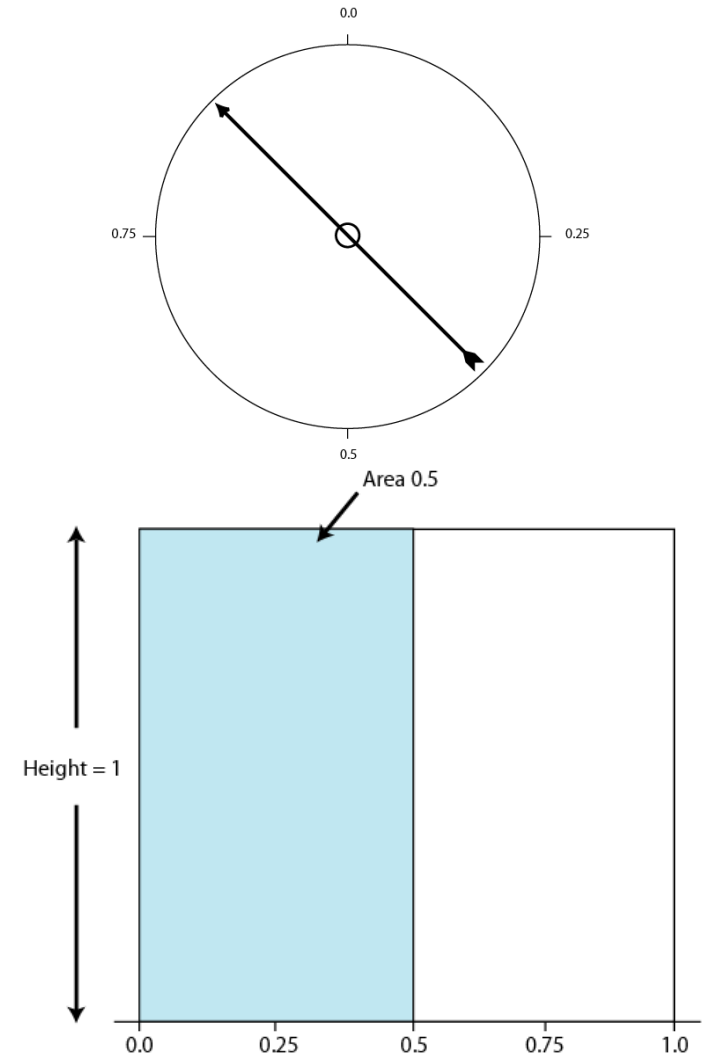


EXAMPLE : CONTINUOUS RANDOM VARIABLE

probability density function (*pdf*) $\equiv$ a mathematical function that assigns probabilities for continuous random variables

The probability of any exact value is 0

BUT, the probability of a **range** is the area under the *pdf* “curve” (bottom)



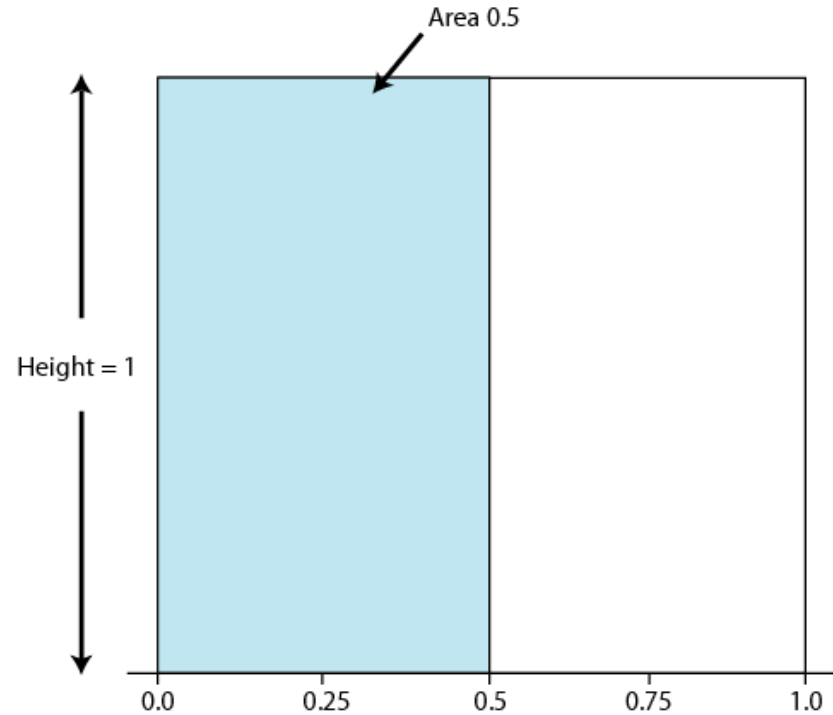
EXAMPLE : CONTINUOUS RANDOM VARIABLE

Area = probabilities

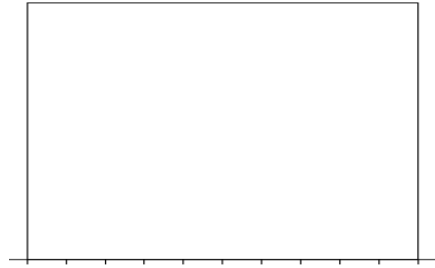
The *pdf* for the random spinner variable $\Rightarrow$

The probability of a value between 0 and 0.5 $\Pr(0 \leq X \leq 0.5)$

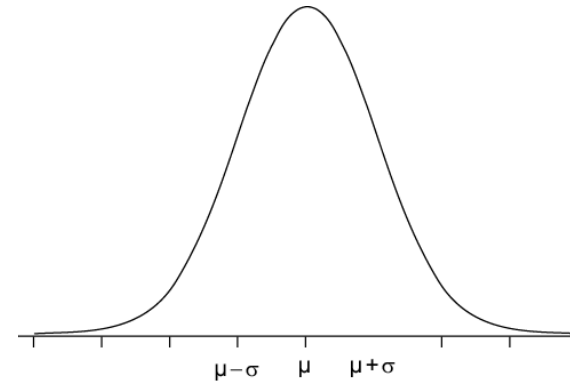
= shaded rectangle
= height $\times$ base
= $1 \times 0.5 = 0.5$



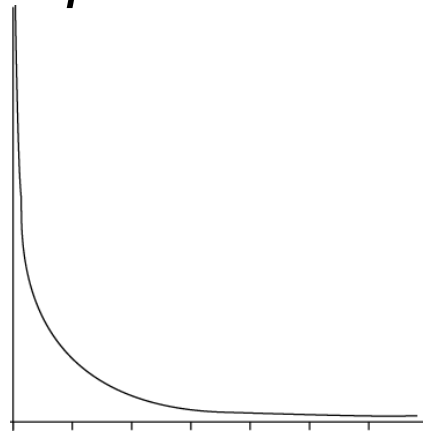
PDF'S COME IN VARIOUS SHAPES HERE ARE EXAMPLES



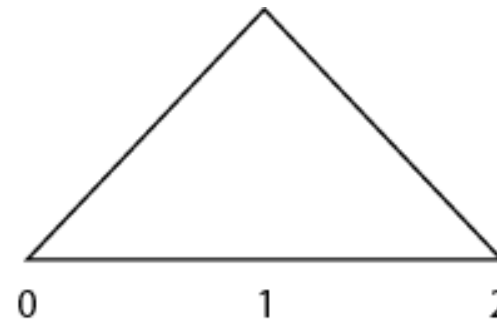
Uniform
pdf



Normal
pdf



Chi-square
pdf



Triangular

ar⁵: Probability Concepts

AREAS UNDER THE C

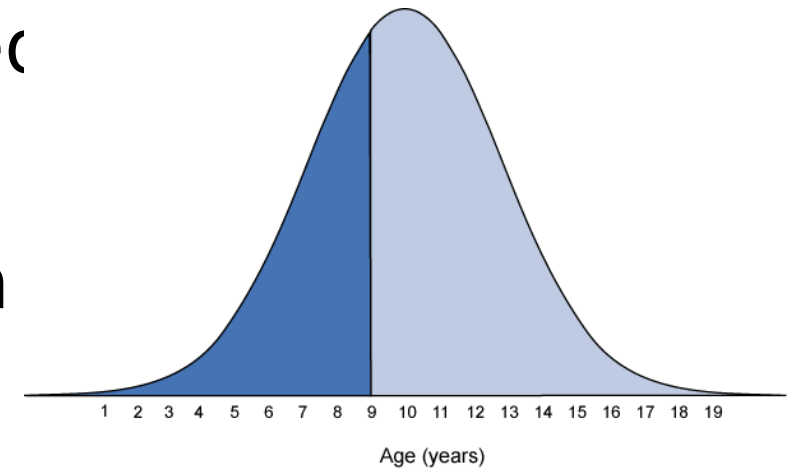
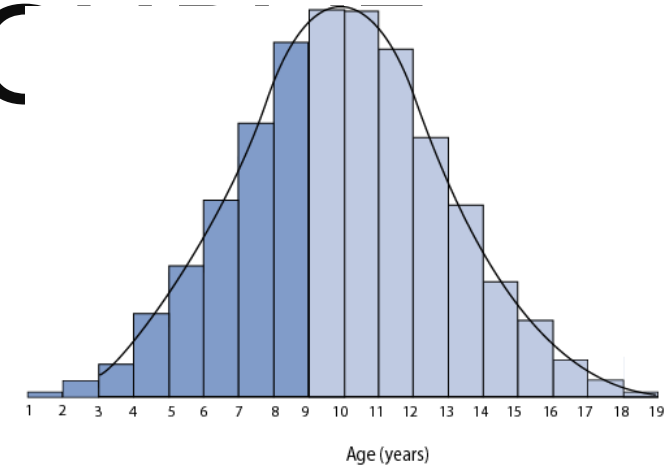
pdf curves are analogous to probability histograms

AREAS = probabilities

Top figure: histogram, ages ≤ 9 shaded

Bottom figure: *pdf*, ages ≤ 9 shaded

Both represent proportion of population
9



PROPERTIES OF PROBABILITIES

Property 1. Probabilities are always between 0 and 1

Property 2. The **sample space (S)** for a random variable represents all possible outcomes and must sum to 1 exactly.

Property 3. The probability of the **complement** of an event (*“NOT the event”*) = 1 MINUS the probability of the event.

Property 4. Probabilities of **disjoint events** can be added.

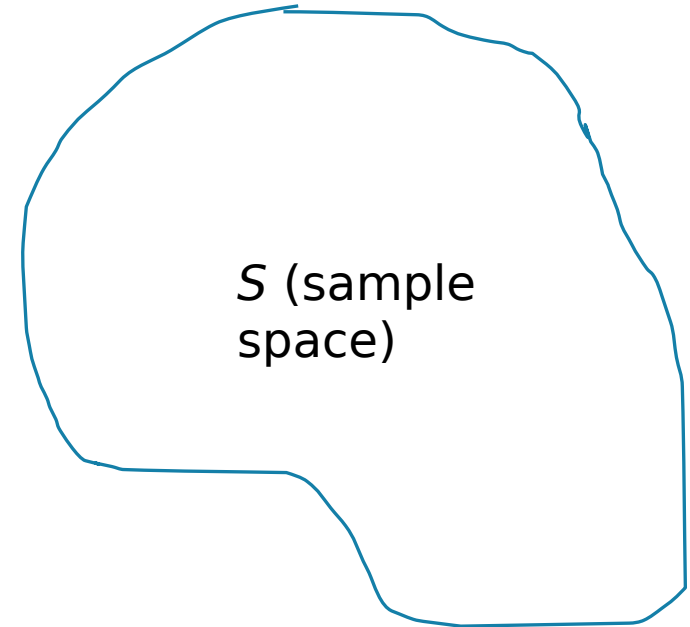
PROPERTIES OF PROBABILITIES IN SYMBOLS

Property 1. $0 \leq \Pr(A) \leq 1$

Property 2. $\Pr(S) = 1$

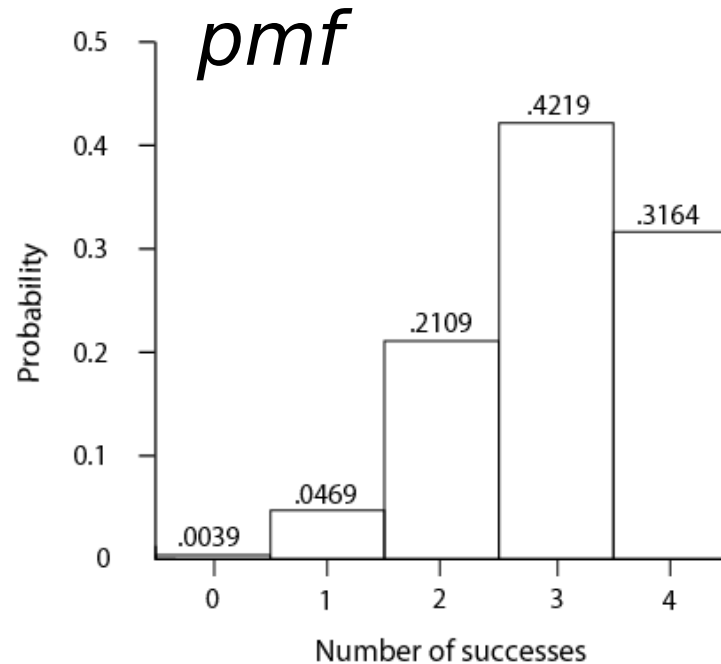
Property 3. $\Pr(\bar{A}) = 1 - \Pr(A)$,
 $\bar{A}$ represents the complement of A

Property 4. $\Pr(A \text{ or } B) = \Pr(A) + \Pr(B)$
when **A and B are disjoint**



PROPERTIES 1 & 2 ILLUSTRATED

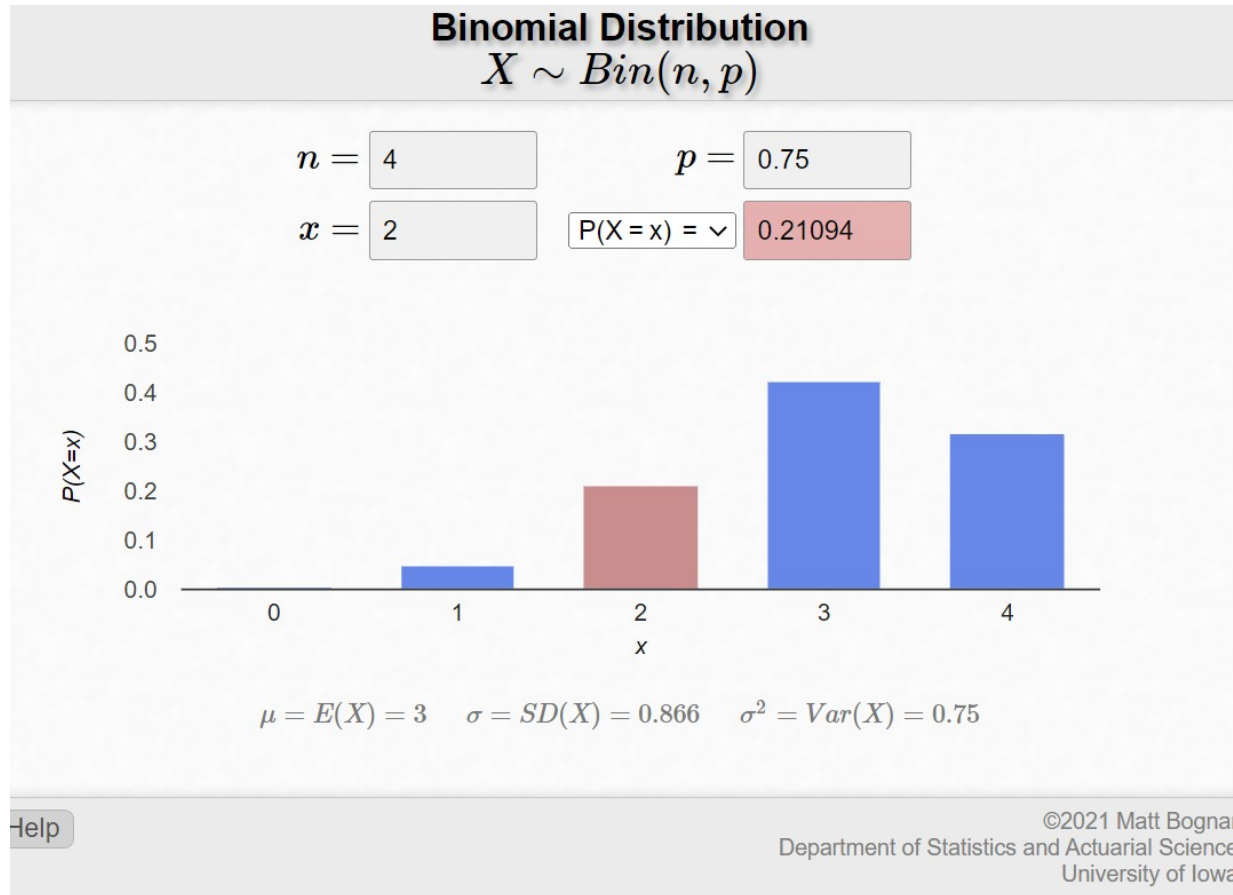
“Four people- literate”



Property 1. Note that all probabilities are between 0 and 1.

Property 2. The sample space sums to 1:
 $\Pr(S) = .0039 + .0469 + .2109 + .4219 + .3164 = 1$

BINOMIAL DISTRIBUTION



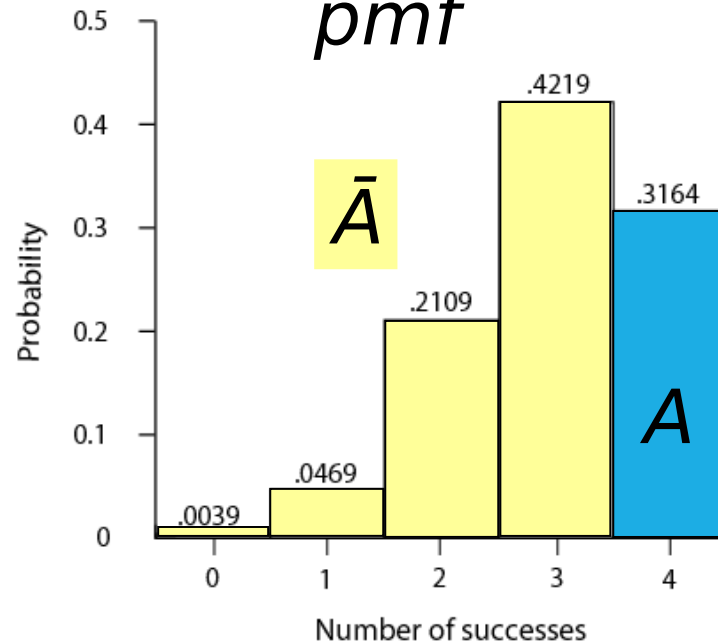
0.0039	0.0469	0.2109	0.4219	0.3164
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<https://homepage.divms.uiowa.edu/~mbognar/applets/bin.html>

PROPERTY 3 (“COMPLEMENTS”)

Let $A \equiv 4$ successes

“Four patients”
pmf



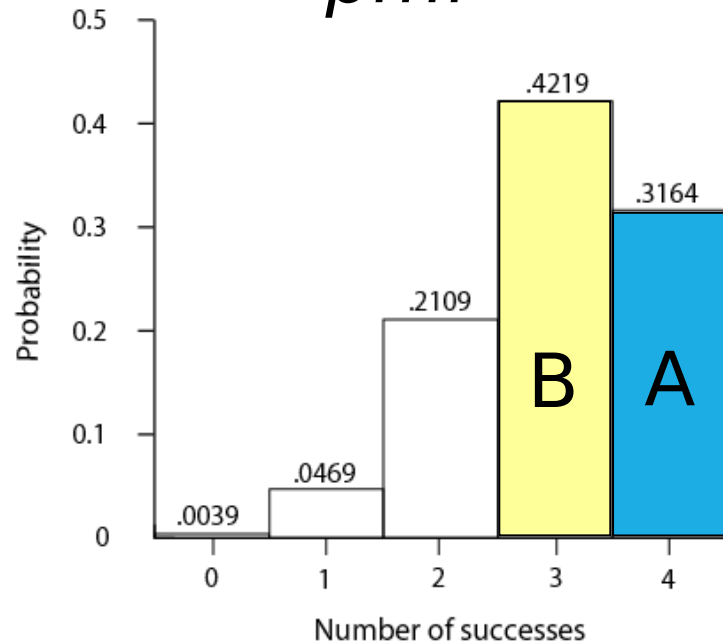
Then, $\bar{A} \equiv$ “not A ” = “**3 or fewer successes**”

Property of complements:

$$\begin{aligned}\Pr(\bar{A}) &= 1 - \Pr(A) \\ &= 1 - 0.3164 \\ &= 0.6836\end{aligned}$$

PROPERTY 4 (DISJOINT EVENTS)

“Four patients”
pmf



Let A represent 4 successes

Let B represent 3 successes

A & B are disjoint

The probability of observing 3 or 4:

$\Pr(A \text{ or } B)$

$= \Pr(A) + \Pr(B)$

$= 0.3164 + 0.4219$

$= 0.7293$

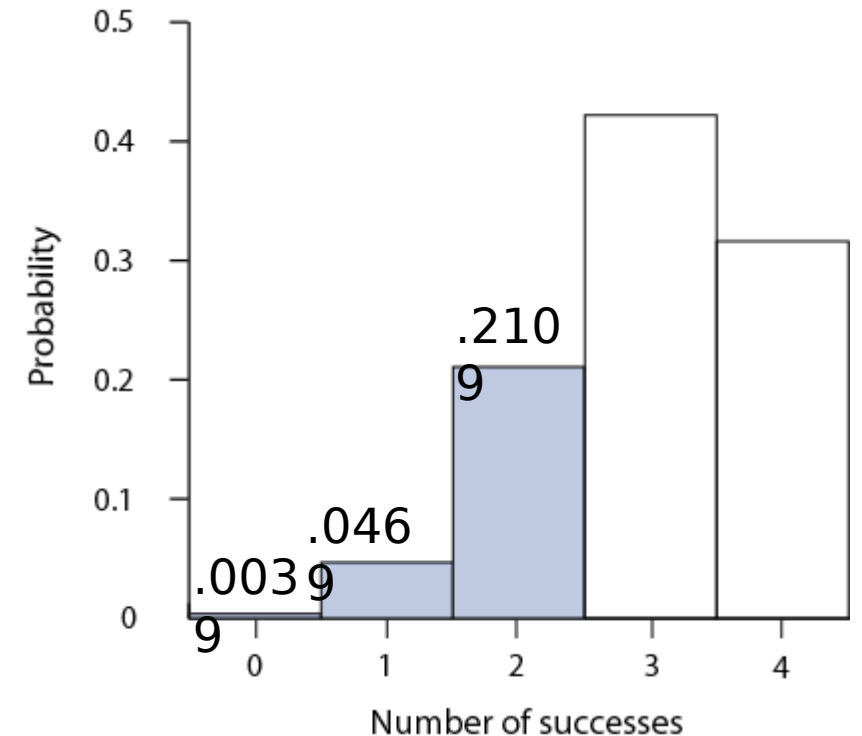
CUMULATIVE PROBABILITY: LEFT "TAIL"

Cumulative probability = probability of x or less

Denoted $\Pr(X \leq x)$

Corresponds to area in **left tail**

Example: $\Pr(X \leq 2)$
= area in left tail
= $.0039 + .0469 + .2109$
= 0.2617



RIGHT “TAIL”

Probabilities greater than a value are denoted **$\Pr(X > x)$**

Complement of cumulative probability

Corresponds to area in **right tail of distribution**

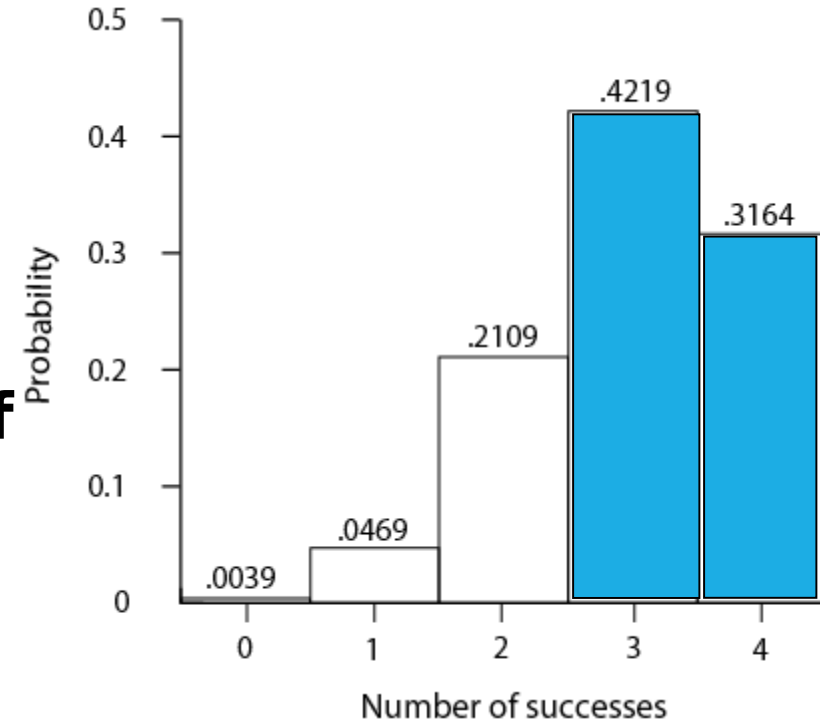
Example (4 patients *pmf*):

$\Pr(X > 3)$

= complement of $\Pr(X \leq 2)$

= $1 - 0.2617$

= .7389



MEAN AND VARIANCE OF A DISCRETE RV

Definitional formulas

The **mean**, i.e.,
expectation

$$\mu = \sum x \cdot \Pr(X = x)$$

The
variance

$$\sigma^2 = \sum (x - \mu)^2 \cdot \Pr(X = x)$$

MOMENTS OF PROBABILITY DISTRIBUTION

- The n^{th} moment of X is represented as $E[X^n]$, where,

$$E[X^n] = \begin{cases} \sum_i x_i^n p(x_i) , & \text{discrete} \\ \int_{-\infty}^{\infty} x^n f(x) dx , & \text{continuous} \end{cases}$$

- The n^{th} central moment of X is represented as $\mu[X^n]$, where,

$$\mu[X^n] = E[(X - \mu)^n] = \begin{cases} \sum_i (x_i - \mu)^n p(x_i) , & \text{discrete} \\ \int_{-\infty}^{\infty} (x - \mu)^n f(x) dx , & \text{continuous} \end{cases}$$

**Central
Moment**

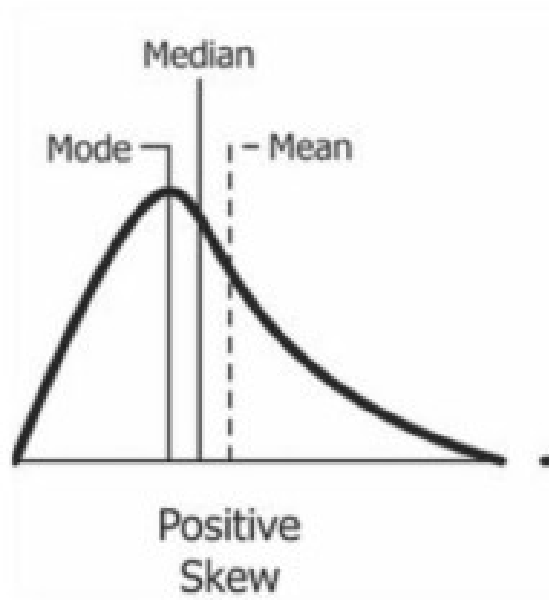
Moment	Uncentered	Centered
1st	$E(X) = \mu$	
2nd	$E(X^2)$	$E((X-\mu)^2)$
3rd	$E(X^3)$	$E((X-\mu)^3)$
4th	$E(X^4)$	$E((X-\mu)^4)$

$$\text{Mean}(X) = E(X)$$

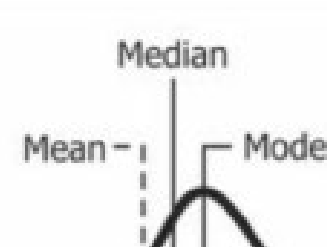
$$\text{Var}(X) = E((X-\mu)^2) = \sigma^2$$

$$\text{Skewness}(X) = E((X-\mu)^3) / \sigma^3$$

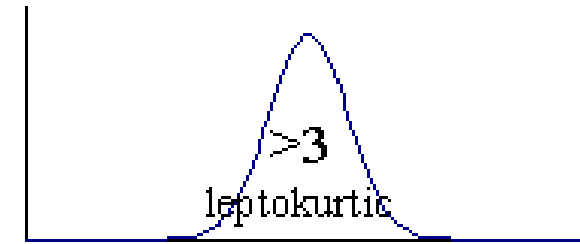
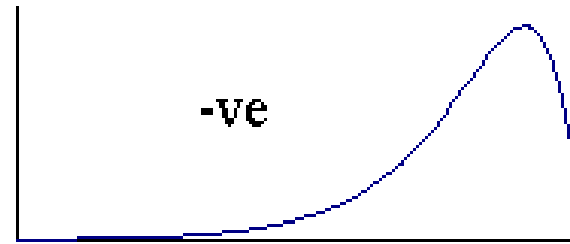
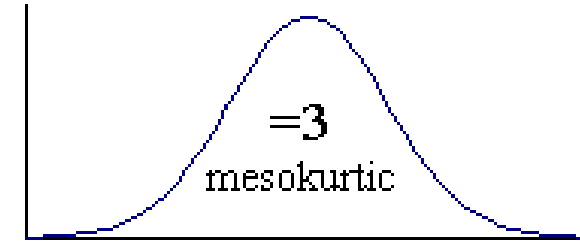
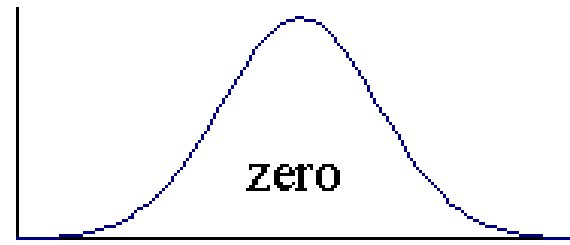
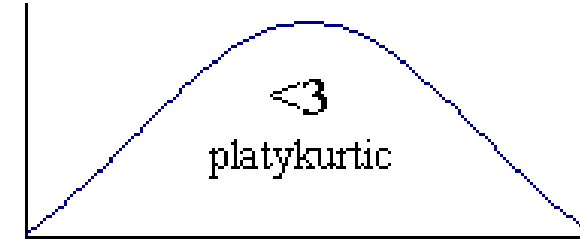
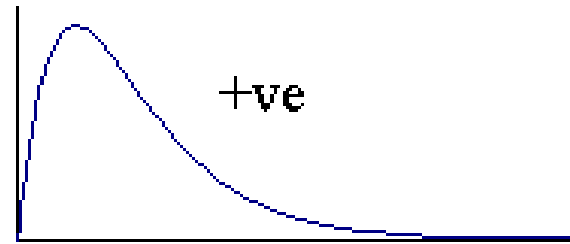
$$\text{Kurtosis}(X) = E((X-\mu)^4) / \sigma^4$$



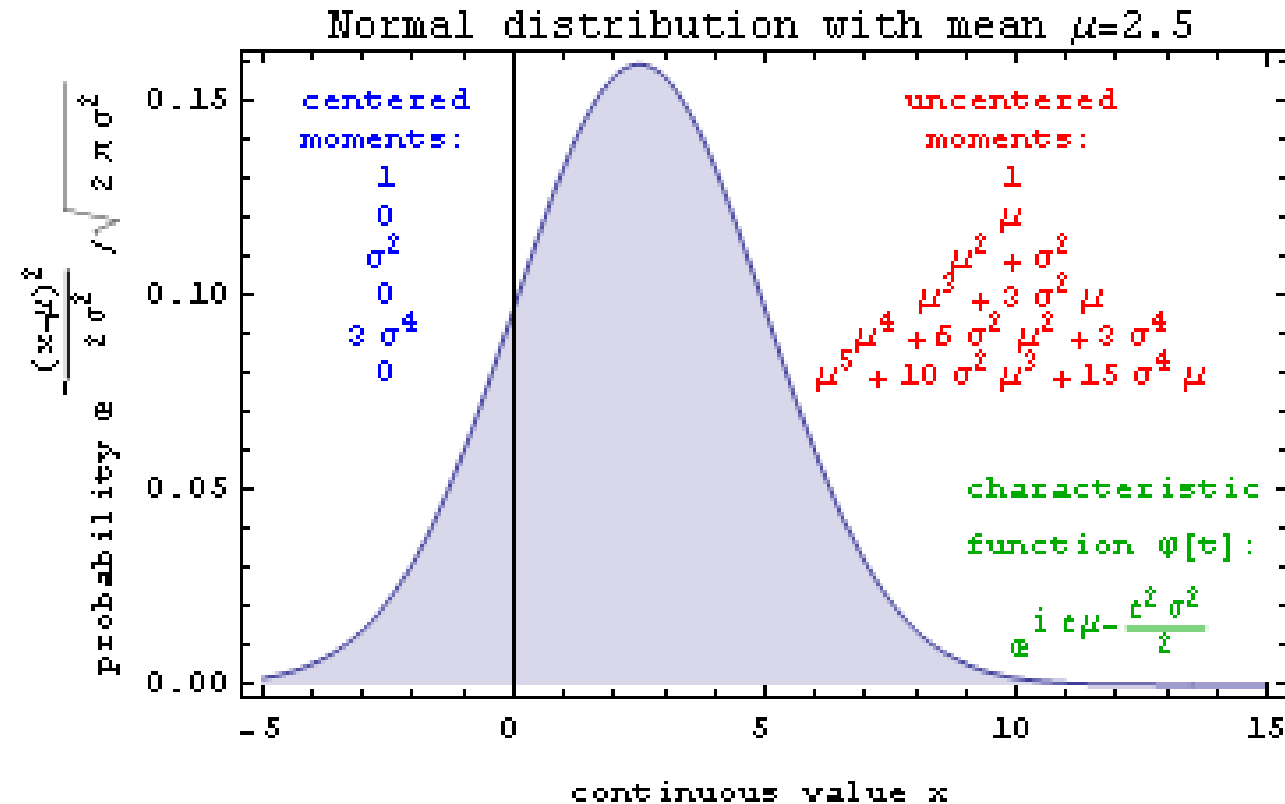
Skewness



Kurtosis



MOMENTS FOR NORMAL DISTRIBUTION



PROBABILITY USING FREQUENCY TABLES

TABLE 4 *Education*

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	Less than 12 Years	781	17.3	17.3	17.3
	12 Years	1,204	26.7	26.7	44.0
	More than 12 Years	2,525	56.0	56.0	100.0
	Total	4,510	100.0	100.0	

Assumption of
**sampling
randomly**

17.3% of the population has less than 12 years of education, 26.7% has 12 years of education, and 56.0% has more than 12 years of education.

The probability of randomly selecting a respondent with less than 12 years of education is $P = .173$.

$P(\text{Completed 12 years of education}) = .267$

$P(\text{Completed more than 12 years of education}) = .56$

EDUCATION COMPLETION VARY BY RACE?

TABLE 5 *Education*

RACE OF RESPONDENT			FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
WHITE	Valid	Less than 12 Years	428	13.0	13.0	13.0
		12 Years	917	27.9	27.9	41.0
		More than 12 Years	1,939	59.0	59.0	100.0
		Total	3,284	100.0	100.0	
BLACK	Valid	Less than 12 Years	137	21.6	21.6	21.6
		12 Years	165	26.0	26.0	47.6
		More than 12 Years	332	52.4	52.4	100.0
		Total	634	100.0	100.0	
OTHER	Valid	Less than 12 Years	216	36.5	36.5	36.5
		12 Years	122	20.6	20.6	57.1
		More than 12 Years	254	42.9	42.9	100.0
		Total	592	100.0	100.0	

$\bar{X} =$
no of years of education completed
(3 groups)

Marital Status

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	MARRIED	2,170	48.1	48.2	48.2
	WIDOWED	366	8.1	8.1	56.3
	DIVORCED	732	16.2	16.3	72.6
	SEPARATED	156	3.5	3.5	76.0
	NEVER MARRIED	1,080	23.9	24.0	100.0
Total		4,504	99.9	100.0	
Missing	NA	6	.1		
Total		4,510	100.0		

1. What is the probability of randomly selecting a divorced respondent?
2. What is the probability of randomly selecting either a married or a widowed respondent?
3. What is the probability of selecting first a separated respondent, then a never married respondent on consecutive turns?
4. What is the probability of selecting, on consecutive picks, first either a widowed or a divorced respondent and second a separated respondent?